

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12399044
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Bergeron; Joe

Diagnostic coverage improvements in redundant sensor element fault detection architectures

Abstract

Aspects of the present disclosure provide apparatuses for minimizing the impacts of differences in sensor measurements while sensors are being used. An apparatus in accordance with an aspect of the present disclosure may comprise a first sensor, a second sensor, a combiner, coupled to the first sensor and the second sensor, and a comparator, coupled to the first sensor and the second sensor, wherein the combiner and comparator modify a dynamic limit of the apparatus based at least in part on a first uncertainty in a first output of the first sensor and a second uncertainty in a second output of the second sensor.

Inventors:	Bergeron; Joe (Greensboro, NC)
Applicant:	Analog Devices, Inc. (Wilmington, MA)
Family ID:	1000008781119
Assignee:	Analog Devices, Inc. (Wilmington, MA)
Appl. No.:	17/985694
Filed:	November 11, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240159578 A1	May. 16, 2024

Publication Classification

Int. Cl.: G01D21/02 (20060101); G01D18/00 (20060101); G06F11/00 (20060101); G05B23/02 (20060101)

U.S. Cl.:

Field of Classification Search

CPC: B60W (20/50); B60W (50/02); B60W (50/0205); B60W (50/023); B60W (50/029); B60W (2050/021); B60W (2050/0215); B60W (2050/0292); B60W (2050/0295); B60W (2050/0297); G05B (9/03); G05B (23/0218); G05B (23/0221); G05B (23/0224); G05B (23/0227); G05B (23/0229); G05B (23/0232); G05B (23/0235); G05B (23/0237); G05B (23/024); G05B (23/0243); G05B (23/0245); G05B (23/0248); G05B (23/0251); G05B (23/0254); G05B (23/0256)

References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5710776	12/1997	Tomlinson	714/704	G05B 9/03
5761090	12/1997	Gross	706/45	G05B 19/4184
5774378	12/1997	Yang	702/109	G01F 15/024
6741919	12/2003	Schuster	701/29.9	G05B 9/02
6757579	12/2003	Pasadyn	700/52	G05B 13/04
11914357	12/2023	Nguyen	N/A	G05B 23/0254
12163867	12/2023	Vilim	N/A	G01M 99/00
2006/0085102	12/2005	Doel	701/4	G05B 23/0232
2009/0198408	12/2008	Salman	701/29.2	B60W 50/0205
2011/0125419	12/2010	Bechhoefer	702/34	F03D 17/00
2013/0293217	12/2012	Moiseev	324/76.77	G05B 9/03
2015/0169393	12/2014	Shibuya	702/182	G05B 23/0243
2015/0351670	12/2014	Vanslyke	600/301	A61B 5/746
2017/0370985	12/2016	Rachmawati	N/A	G05B 9/02
2018/0031428	12/2017	Uno	N/A	G01K 3/005
2018/0307218	12/2017	Lavid Ben Lulu	N/A	G06F 16/90335
2019/0235482	12/2018	Subramaniyan	N/A	G05B 23/0243
2021/0149465	12/2020	Hiltner	N/A	G06F 1/206
2021/0335059	12/2020	Dixit	N/A	G05B 23/0254
2022/0051017	12/2021	Choi	N/A	G06V 10/764
2022/0162998	12/2021	Lee	N/A	F02C 7/32
2022/0171376	12/2021	Kim	N/A	G05B 19/41885
2022/0176547	12/2021	Smith	N/A	A61H 1/0255
2023/0195413	12/2022	Kastrup	708/200	H03K 19/21
2023/0400508	12/2022	Freese	N/A	G05B 23/0235
2024/0308660	12/2023	Strihagen	N/A	B64D 7/00

Primary Examiner: Roberts; Herbert K

Attorney, Agent or Firm: ArentFox Schiff LLP

Background/Summary

BACKGROUND

Technical Field

(1) The present disclosure generally relates to fault detection, and more particularly, to diagnostic coverage improvements in redundant sensor element fault detection architectures.

Introduction

(2) Electronic devices are often used as sensors that respond to physical stimuli, and other electronics, broadly referred to as transducers, provide an electronic representation of the sensor's response. Within some systems, multiple sensors may be used to provide enhanced sensor responses. The response from each sensor may be averaged or otherwise compared to allow for better performance, reduced errors, identifying faults, or improving other system parameters.

(3) There are also many possibilities for errors from sensors in redundant sensor systems. Differences in operating temperature, power supplied to each sensor, signal levels, etc., may provide different responses that are then compared, averaged, etc., to achieve an electronic representation of the overall response. Reducing these inaccuracies in each sensor or each of the sensor paths may increase costs. Further, operating sensors away from nominal operating points may increase the errors that sensors introduce, which may degrade the specifications for a given sensor to accommodate the additional range of infrequent but expected larger inputs.

SUMMARY

(4) The following presents a simplified summary of one or more aspects in order to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated aspects, and is intended to neither identify key or critical elements of all aspects nor delineate the scope of any or all aspects. Its sole purpose is to present some concepts of one or more aspects in a simplified form as a prelude to the more detailed description that is presented later.

(5) The present disclosure provides apparatuses for minimizing the impacts of differences in sensor measurements while sensors are being used. In an aspect of the present disclosure, sensors can be operated at a wider range of conditions. The present disclosure provides apparatuses for identifying sensor faults with increased resolution, even with very wide range sensors that could otherwise mask faults.

(6) An apparatus in accordance with an aspect of the present disclosure may comprise a first sensor, a second sensor, a combiner, coupled to the first sensor and the second sensor, and a comparator, coupled to the first sensor and the second sensor, wherein the combiner and comparator modify a dynamic limit of the apparatus based at least in part on a first uncertainty in a first output of the first sensor and a second uncertainty in a second output of the second sensor. In an aspect of the present disclosure, such an apparatus may enable increased resolution of faults that can be detected in spite of using wider range sensors.

(7) Such an apparatus further optionally includes the combiner averaging the first output and the second output, the comparator comparing an average of the first output and the second output against a threshold, the first and/or second uncertainty respectively in the first output and/or the second output being determined at least in part by one or more of a noise error, an uncertainty error, a scale error, a quantization error, and a sensor uncertainty, the dynamic limit being based at least in part on an input to the first sensor and the second sensor, and the dynamic limit increasing as the input deviates from a nominal value.

(8) Such an apparatus further optionally comprises a first averager coupled to the first sensor, the first averager averaging a plurality of first outputs of the first sensor, the combiner averaging an output of the first averager with the second output, a second averager coupled to the second sensor, the second averager averaging a plurality of second outputs of the second sensor, and the combiner averaging an output of the first averager and an output of the second averager. Averaging or filtering sensor responses may offer an opportunity for increased fidelity in fault detection.

(9) An integrated sensor in accordance with an aspect of the present disclosure may comprise a first

sensor, a second sensor, a combiner, coupled to the first sensor and the second sensor, and a comparator, coupled to the first sensor and the second sensor, wherein the combiner and comparator modify a dynamic limit of the integrated sensor based at least in part on an average of a first uncertainty in a first output of the first sensor and a second uncertainty in a second output of the second sensor.

(10) Such an integrated sensor may further optionally include the combiner averaging the first output and the second output, the comparator comparing an average of the first output and the second output against a threshold, the first and/or second uncertainty respectively in the first output being determined at least in part by one or more of a noise error, an uncertainty error, a scale error, a quantization error, and a sensor uncertainty, and the dynamic limit being based at least in part on an input to the first sensor and the second sensor.

(11) Such a sensor arrangement may further optionally include a first averager coupled to the first sensor, and the combiner averages an output of the first averager.

(12) To the accomplishment of the foregoing and related ends, the one or more aspects comprise the features hereinafter fully described and particularly pointed out in the claims. The following description and the annexed drawings set forth in detail certain illustrative features of the one or more aspects. These features are indicative, however, of but a few of the various ways in which the principles of various aspects may be employed, and this description is intended to include all such aspects and their equivalents.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram of an integrated sensor in accordance with an exemplary aspect of the present disclosure.

(2) FIG. 2 is a diagram of the scale error impact of sensor uncertainty and faults in accordance with an exemplary aspect of the present disclosure.

(3) FIG. 3 is a graph of dynamic limits versus input value in accordance with an exemplary aspect of the present disclosure.

(4) FIG. 4 is a block diagram of a non-dynamic comparison of redundant sensors in accordance with an exemplary aspect of the present disclosure.

(5) FIG. 5 is a block diagram of a dynamic comparison of redundant sensors in accordance with an exemplary aspect of the present disclosure.

(6) FIG. 6 is a block diagram of a dynamic comparison of redundant sensors in accordance with an exemplary aspect of the present disclosure.

DETAILED DESCRIPTION

(7) The detailed description set forth below in connection with the appended drawings is intended as a description of various configurations and is not intended to represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of various concepts. However, it will be apparent to those skilled in the art that these concepts may be practiced without these specific details. In some instances, well known structures and components are shown in block diagram form in order to avoid obscuring such concepts.

Introduction

(8) For a system to operate properly and safely, determination of system safety goals and their allocation to items and elements in the system may be used to reduce opportunities for unsafe faults. Faults within a system may be defined as conditions that can cause termination of the system's ability to perform its function as desired. When faults are detected, action can be taken for the system to remain in a safe operating state. Alternatively, or in addition to fault detection, faults

can be restricted or categorized to levels appropriate for the risks introduced by the fault and/or the probability of occurrence of such faults.

(9) Because many elements and sensors are often used in designs, there can be difficulties in ensuring individual sensors (also referred to as “elements” herein) avoid developing undetected random faults and failures. Random faults and failures may lead to undetected safety goal violations. As such, sensor implementation may be designed to manage and limit these random faults to levels commensurate with the risks and exposure that they can create. Alternatively, or in addition to this limitation of safety risk, the system or item using the sensor may adopt alternative approaches to determine faults to manage the faults appropriately.

(10) Many systems attempt to improve performance assessments using a redundant sensor element fault detection architecture to aide in assuring functional safety goal fulfilment. Using sensor measurement statistics and employing measurement dependent redundancy expectations to assess the sensors' performance can improve error resolution without increased false alarm reports. Using increased sensor performance expectations in lieu of signal independent defined sensor performance can improve system capability and increase safety performance goals.

(11) Often, diagnostic safety mechanisms (DSMs) identify fault outcomes and limit system exposure to potential failure. DSMs can reveal abnormal performance by assessing symptomatic responses, as well as monitor other sensor behaviors to measure operational changes or outputs beyond design expectations. Using DSMs to meet a safety goal often employs testing sensors against the sensor's operational limits.

(12) In general, an element or sensor's suitability for a given application, and performance in a given application can be evaluated, and its capabilities assessed for their ability to meet functional safety requirements.

Overview

(13) FIG. 1 is a block diagram of an integrated sensor in accordance with an exemplary aspect of the present disclosure.

(14) FIG. 1 illustrates an integrated sensor **100** (which may also be referred to as an apparatus herein), having a stimulus **102** as an input and also having an output **104**. Stimulus **102** may be a force, an acceleration, or other stimulus that can be detected by integrated sensor **100**, while output **104** may be an electrical, optical, or other signal that can be transmitted or interpreted by another system or directly by a user. Within integrated sensor **100** is a sensor response **106**, a sampler **108**, a transducer **110**, an adder **112**, a scaler **114**, a quantizer **116**, an output response **118**, a digitizer **120**, and a calibrator **122**. Other block diagrams that describe similar functions for integrated sensor **100** may be used without departing from the scope of the present disclosure.

(15) Sensor response **106** is a representation of the sensor response curve for various stimuli. For example, and not by way of limitation, sensor response **106** may be the frequency response of a given sensor, the sensitivity response of a gyro, etc. Sampler **108** is a representation of how often the sensor element is sampled, e.g., once a second, once every millisecond, etc.

(16) Transducer **110** converts and/or amplifies the sampled sensor response to an electrical signal that can be scaled and calibrated. Adder **112** allows for the addition of bias signal(s) **124**, but also shows that noise **126** may be added into the electrical signal coming from the sensor. Noise **126** may include, e.g., cross-sensor contamination, electromagnetic interference, operational conditions such as vibration of the sensors, and/or other inputs that become part of the sensor output **104** not generated by the elements of the integrated sensor **100**.

(17) Scaler **114** amplifies the output of adder **112**, which is then input to quantizer **116**. Quantizer **116**, which is coupled to digitizer **120**, may allow for a digitized output from quantizer **116**. Output response **118**, which is coupled to calibrator **122**, may provide output **104** in a calibrated manner.

(18) Within integrated sensor **100**, sensor response **106**, sampler **108**, transducer **110**, adder **112**, scaler **114**, quantizer **116**, and output response **118** may each contribute to errors and/or faults that can be reported by the sensor. These errors and/or faults can be linear or non-linear in nature.

Further, these errors and/or faults may be repeatable for a certain stimulus **102**, may be a degradation of one or more portions of integrated sensor **100** over the life of the integrated sensor **100**, or may result from a failure of one or more portions of the integrated sensor **100**. To assist in assessing the performance and applicability of the integrated sensor, various approaches are used to ensure increased quality and safety goals.

(19) FIG. **2** is a diagram of the scale error impact of sensor uncertainty and faults in accordance with an exemplary aspect of the present disclosure.

(20) Graph **200** illustrates time **202** on a horizontal (x) axis and quantization **204** on a vertical (y) axis, with an input **206** to a sensor. The reported measurements **208** are shown as solid dots, with noise **210** and uncertainty **212** of each of the reported measurements **208**. The scale error **214** is shown as dotted and dashed lines around the input **206**. The quantization **204** of a given measurement, having a sensor uncertainty **216**, is also shown along the quantization **204** axis.

(21) For each of the reported measurements **208** of input **206**, FIG. **2** shows that there are various error components, e.g., the noise **210**, uncertainty **212**, scale error **214**, and sensor uncertainty **216** that can lead to an output from the sensor that is different than the input **206**. Once these errors add up to a value that falls outside of the sensor uncertainty **216**, a fault **218** may be reported. For each sensor used in an integrated sensor, the error components, e.g., the noise **210**, uncertainty **212**, scale error **214**, and sensor uncertainty **216** can be different, as the sensors in a redundant and/or integrated sensor are physically different devices or components.

(22) The block diagram of FIG. **1** may be linked to the output shown in FIG. **2**. For example, noise **210** of FIG. **2** may be represented by noise **126** in FIG. **1**, quantization **126** may be represented by quantization **204**, stimulus **102** may be represented by input **206**, etc.

(23) FIG. **3** is a graph of dynamic limits versus input value in accordance with an exemplary aspect of the present disclosure.

(24) Graph **300** shows input value **302**, noise error **304**, bias error **306**, scale error **308**, dynamic limit **310**, and nominal operating point **312**. As with FIGS. **1** and **2**, various portions of graph **300** may be illustrated or represented in FIG. **3**, e.g., input value **302** may be the stimulus **102** and/or input **206**, noise error **304** may be illustrated by noise **126** or noise **210**, etc.

(25) Input value **302** represents the input value, e.g., stimulus **102**, received by the sensor, e.g., integrated sensor **100**. Input value **302** can be received by the sensor at any point on the full-scale range (FSR) of the sensor, including at the nominal operating point **312**.

(26) Noise error **304** is a representation of the noise and inherent differences of output for a given reading by the sensor. As noise, e.g., noise **126** shown in FIG. **1**, is introduced into the sensor, the reading may be affected.

(27) Bias error **306** represents errors in the bias of the sensor, e.g., bias **124** as shown in FIG. **1**.

(28) Scale error **308** may indicate the difference in sensor part performance. For example, two sensors with a 1% scale factor error and a 500 unit (u) full scale range (FSR) would have a potential difference in readings of $0.01 * (500u - (-500u)) = 10u$.

(29) The 500u FSR could be reduced; however, reducing the FSR increases the potential exposure to inputs beyond the range of this reduced FSR, which may limit the applications that a given sensor can be operated in. Further, maintaining a larger FSR allows for sensor characteristics that may be desired in the application, e.g., faster recovery from transients, integration of transients for system response, etc. A scale error **308** of 1% may be helpful in detecting failures, but there may be tolerance gaps between the performance expectations and the safety concept requirements whenever the scale error **308** is beyond the intended functionality expectations.

(30) Dynamic limit **310** represents the limit of performance response for a given sensor about the nominal operating point **312**. As inputs deviate from the nominal operating point **312**, the uncertainty of the output of the sensor increases. For example, at nominal operating point **312**, the dynamic limit **310** is shown as limit value **314**. When the input occurs at a different operating point, an exemplary limit value is shown as limit value **316**.

(31) The dynamic limit **310** can be described as the sum of the noise error **304**, bias error **306**, and scale error **308** at a given operating point. Because the sensor is designed to operate about nominal operating point **312**, the dynamic limit **310** may be greater as operating points for the sensor approach the FSR of the sensor, as shown by limit value **314**. At such operating points, the possible errors of the output of the sensor are greater. As seen in FIG. 3, the dynamic limit **310** may increase the ability to detect faults with an increase in input value to the sensor.

(32) The dynamic limit **310** may also be described as the limit on the difference between sensors, e.g., differences caused by expected sensor matching and faults. Although dynamic limit **310** may not be increased in aspects of the present disclosure, modification of dynamic limit **310** provides opportunities to increase fault detection are enabled by reducing difference expectations in the dynamic limit **310**.

(33) FIG. 4 is a block diagram of a non-dynamic comparison of redundant sensors in accordance with an exemplary aspect of the present disclosure.

(34) Many applications require sensor ranges that minimize the opportunities for saturation of the sensor during use. This requirement creates a full scale range that may be much greater than the nominal range of input values. Errors associated with scale may be the main consideration when developing expectations for a multi-sensor comparison function. In an aspect of the present disclosure, increasing the dynamic limit for signals that are outside of mid-scale reading may provide an increase in performance for an integrated sensor system.

(35) System **400** (which may also be referred to as an apparatus herein) may have an input **402**, and may include sensor **1 404**, sensor **2 406**, and comparator **408**. The output of system **400** may be designated as output **410**, while the output of comparator **408** may be designated as comparator output **412**.

(36) As an example, and not by way of limitation, sensor **1 404** may have a fault indication, while sensor **2 406** does not have a fault indication. In such an example, the output **410** of sensor **1 404** ("S1 Output") would be:

$$S1\text{ Output} = \text{Input} + S1(\text{uncertainty}) + \text{fault value}$$

(37) The output of sensor **2 406** ("S2 Output") would be:

$$S2\text{ Output} = \text{Input} + S2(\text{uncertainty})$$

(38) Where the uncertainties of each sensor would be:

$$S1(\text{uncertainty}) \leq ns1 + os1 + SE1 * In(\text{max})$$

$$S2(\text{uncertainty}) \leq ns2 + os2 + SE2 * In(\text{max})$$

(39) Where ns is the noise error **304**, os is the bias error **306**, and SE is the scale error **308**, while $In(\text{max})$ is the FSR of the sensor. Without comparator **408**, system **400** could not determine whether output **410** had a fault indication or not.

(40) Comparator **408** compares the outputs, e.g., S1 Output (i.e., output **410**) and S2 Output. Of course, if more sensors are included in system **400**, then comparator **408** may compare some or all of the outputs from each of the sensors included. In the example of FIG. 4, comparator **408** may subtract the outputs of sensor **1 404** and sensor **2 406**, and may take the absolute value of this result if desired. As such, the comparator output **412** of system **400** may be:

$$\text{Output} = S1(\text{uncertainty}) - S2(\text{uncertainty}) - \text{fault value}$$

(41) Which, upon substitution, yields:

$$\text{Output} \geq 2 * (ns(\text{max}) + os(\text{max}) + SE(\text{max}) * FS)$$

(42) Where $ns(\text{max})$ is the maximum noise error **304**, $os(\text{max})$ is the maximum bias error **306**, and $SE(\text{max})$ is the maximum scale error **308**.

(43) This comparator output **412** can then be assessed against a diagnostic limit to determine if output **410** contains a fault indication.

(44) FIG. 5 is a block diagram of a dynamic comparison of redundant sensors in accordance with an exemplary aspect of the present disclosure.

(45) System **500** (which may also be referred to as an apparatus herein) may have an input **502**, and

may include sensor **1 504**, sensor **2 506**, combiner **508**, and comparator **510**. The output of system **500** may be designated as output **512**, while the output of comparator **510** may be designated as comparator output **514**.

(46) In system **500**, combiner **508** may combine the outputs of sensor **1 504** and sensor **2 506** in various ways. For example, and not by way of limitation, combiner **510** may average the outputs of sensor **1 504** and sensor **2 506**, i.e., output **512** may be determined as ((Output of sensor **1 504**+output of sensor **2 506**)/2). Other combinations of the outputs of sensor **1 504** and sensor **2 506**, e.g., a weighted average, etc., are also possible without departing from the scope of the present disclosure.

(47) When comparing the outputs of sensor **1 504** and sensor **2 506**, using output **512** as a comparison for comparator **510** now increases the fault limit for a given integrated sensor:

$$\text{Output} \geq 2 * (ns(\text{max}) + os(\text{max}) + SE(\text{max}) * \text{Input}) + \text{DELTA}$$

(48) Where DELTA equals the (uncertainty of sensor **1 504**+uncertainty of sensor **2 506**+fault)/2. In other words, the difference value to show a fault for the integrated sensor has been increased by the average of the sum of the uncertainties plus the fault reading. The faults modify the dynamic limits by the scale factor times the fault value divided by two, while the fault difference may only be increased by the fault itself. Thus, an integrated sensor now has a broader application, as the sensor can operate at wider ranges away from center of full scale readings, as the fault limits have been expanded to account for modified dynamic operating input ranges. To reduce false fault detection the various errors discussed with respect to FIGS. **1** and **2**, e.g., noise, bias, linearity, timing, quantization, and environmental sensitivity error expectations, may be accounted for when setting comparison limits for comparator **510**.

(49) Further, comparator output **514** may be compared against a threshold or other value to modify a dynamic limit of a given system **500**. For example, and not by way of limitation, the comparator output may be compared against a threshold that is based at least in part on an uncertainty in the output of sensor **1 504** and/or an uncertainty in the output of sensor **2 506**. The uncertainty of the outputs of sensor **1 504** and/or sensor **2 506** may be based at least in part by one or more of an input to the sensor, a noise error, an uncertainty error, a scale error, a quantization error, and a sensor uncertainty as described with respect to FIGS. **1** and **2**.

(50) FIG. **6** is a block diagram of a dynamic comparison of redundant sensors in accordance with an exemplary aspect of the present disclosure.

(51) System **600** (which may also be referred to as an apparatus herein) may have an input **602**, and may include sensor **1 604**, sensor **2 606**, combiner **608**, and comparator **610**. The output of system **600** may be designated as output **612**, while the output of comparator **610** may be designated as comparator output **614**. System **600** also includes combiner **616** and combiner **618**.

(52) As with system **500** discussed with respect to FIG. **5**, combiner **608** may combine the outputs of sensor **1 604** and sensor **2 606** in various ways. For example, and not by way of limitation, combiner **608** may average the outputs of sensor **1 604** and sensor **2 606**, i.e., output **612** may be determined as ((Output of sensor **1 604**+output of sensor **2 606**)/2). Other combinations of the outputs of sensor **1 604** and sensor **2 606**, e.g., a weighted average, etc., are also possible without departing from the scope of the present disclosure.

(53) In addition, sensor **1 604** may also have a combiner **616** (also referred to as an averager **616**) connected to the output of sensor **1 604**, and sensor **2 606** may also have a combiner **618** (also referred to as an averager **618**) connected to the output of sensor **2 606**. Combiner **616** and combiner **618** may take a number of readings from their respective sensors and average those sensor outputs prior to having the outputs of combiners **616** and **618** used as inputs to combiner **608**. This may further broaden the dynamic limits of a given integrated sensor, e.g., system **600**, for a wider range of input values from input **602**. Further, system **600** may provide a way to provide weighted averages for sensor **1 604** and sensor **2 606**, as a larger or smaller number of readings can be averaged by one of combiner **616** and combiner **618** prior to their input to combiner **608**. To

reduce false fault detection the various errors discussed with respect to FIGS. 1 and 2, e.g., noise, bias, linearity, timing, quantization, and environmental sensitivity error expectations, may be accounted for when setting comparison limits for comparator 610.

(54) As the number of samples averaged by combiners 616 and 618 increase, the increased resolution of system 600 may allow the diagnostic coverage of system 600 to increase as well. The sensitivity to single measurement faults reported by system 600 may decrease, as single or low-frequency fault measurements are diffused by the time-averaging and redundant measurement averaging performed by system 600. The impact of a single or low frequency fault may be reduced because such faults have lower impacts the output 612.

(55) Further, and not by way of limitation, the comparator output 614 may be compared against a threshold that is based at least in part on an uncertainty in the output of sensor 1 604 and/or an uncertainty in the output of sensor 2 606. The uncertainty of the outputs of sensor 1 604 and/or sensor 2 606 may be based at least in part by one or more of an input to the sensor, a noise error, an uncertainty error, a scale error, a quantization error, and a sensor uncertainty as described with respect to FIGS. 1 and 2.

(56) The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to the exemplary aspects and aspects presented throughout this disclosure will be readily apparent to those skilled in the art, and the concepts disclosed herein may be applied in other contexts and for different purposes. Thus, the claims are not intended to be limited to the exemplary aspects presented throughout the disclosure, but are to be accorded the full scope consistent with the language claims. All structural and functional equivalents to the elements of the exemplary aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. No claim element is to be construed under the provisions of 35 U.S.C. § 112(f), or analogous law in applicable jurisdictions, unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for.”

Claims

1. An apparatus, comprising: a first sensor configured to output a first output; a second sensor configured to output a second output; a combiner, coupled to the first sensor and the second sensor; and a comparator, coupled to the first sensor and the second sensor, wherein the combiner and comparator are mutually configured to modify a dynamic limit of the apparatus based at least in part on a first uncertainty in the first output of the first sensor and a second uncertainty in the second output of the second sensor, by averaging the first output and the second output, and adding one or more faults detected in the first uncertainty and the second uncertainty, to increase an operating range for detecting uncertainties outputted by the first sensor and/or the second sensor.
2. The apparatus of claim 1, wherein the comparator compares an average of the first output and the second output against a threshold.
3. The apparatus of claim 2, wherein the first uncertainty in the first output is determined at least in part by one or more of a first noise error, a first uncertainty error, a first scale error, a first quantization error, and a first sensor uncertainty.
4. The apparatus of claim 3, wherein the second uncertainty in the second output is determined at least in part by one or more of a second noise error, a second uncertainty error, a second scale error, a second quantization error, and a second sensor uncertainty.
5. The apparatus of claim 4, wherein the dynamic limit is based at least in part on an input to the first sensor and the second sensor.
6. The apparatus of claim 5, wherein the dynamic limit is modified as the input deviates from a

nominal value.

7. The apparatus of claim 1, further comprising a first averager coupled to the first sensor.

8. The apparatus of claim 7, wherein the first averager averages a plurality of first outputs of the first sensor.

9. The apparatus of claim 8, wherein the combiner averages an output of the first averager.

10. The apparatus of claim 7, further comprising a second averager coupled to the second sensor.

11. The apparatus of claim 10, wherein the second averager averages a plurality of second outputs of the second sensor.

12. The apparatus of claim 11, wherein the combiner averages an output of the first averager and an output of the second averager.

13. An integrated sensor, comprising: a first sensor configured to output a first output; a second sensor configured to output a second output; a combiner, coupled to the first sensor and the second sensor; and a comparator, coupled to the first sensor and the second sensor, wherein the combiner and comparator are mutually configured to modify a dynamic limit of the apparatus based at least in part on a first uncertainty in the first output of the first sensor and a second uncertainty in the second output of the second sensor, by averaging the first output and the second output, and adding one or more faults detected in the first uncertainty and the second uncertainty, to increase an operating range for detecting uncertainties outputted by the first sensor and/or the second sensor.

14. The integrated sensor of claim 13, wherein the comparator compares an average of the first output and the second output against a threshold.

15. The integrated sensor of claim 14, wherein the first uncertainty in the first output is determined at least in part by one or more of a noise error, an uncertainty error, a scale error, a quantization error, and a sensor uncertainty.

16. The integrated sensor of claim 13, wherein the dynamic limit is based at least in part on an input to the first sensor and the second sensor.

17. The integrated sensor of claim 13, further comprising a first averager coupled to the first sensor.

18. The integrated sensor of claim 17, wherein the combiner averages an output of the first averager.
